

Hierarchical Risk Parity Variants for Cryptocurrency Portfolios: Denoising, Detoning, Tail Dependence, Embeddings, and Honest Inference Over 2020–2026

Alan Matys¹ and Federico Martin Rodriguez¹

¹Universidad del CEMA , alan.matys92@gmail.com,
federico.m.rodriguez.h@gmail.com

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Abstract

We extend the MACI 2025 study of Hierarchical Risk Parity (HRP) for cryptocurrency portfolios. We close the two future-work items of that paper — Marchenko–Pastur denoising and spectral detoning of the correlation matrix — and broaden the comparison to 19 portfolio strategies: twelve HRP-family variants (the baseline; seven correlation/covariance constructions — detoned, partial-correlation, EWMA-dynamic, lower-tail-dependence, Ledoit–Wolf shrunk-covariance, vol-standardised, and a tail-dependence+shrinkage hybrid; and four embedding-based variants that learn the clustering distance from path signatures, a node2vec graph embedding, and two contrastive sequence encoders), four risk-based comparators (IVP, MVP, ERC, Maximum Diversification), and three crypto-momentum strategies. Every comparison runs on a survivorship-bias-corrected, point-in-time universe rebuilt monthly from Binance USDT quote-volume rankings (146 ever-included symbols, including delisted assets such as LUNA and FTT), uses a 365-day rolling estimation window, and is wrapped in Ledoit–Wolf robust Sharpe-difference, Hansen Superior Predictive Ability, and stationary block-bootstrap inference. On 76 monthly rebalances over 2020-02 to 2026-05, the twelve HRP-family variants bunch in a narrow annualised-Sharpe band (0.70–0.75); the best, HRP_{Dynamic}, exceeds baseline HRP by only +0.008 Sharpe (Ledoit–Wolf $p = 0.82$, not significant). Hansen SPA gives $p_{\text{consistent}} = 0.46$ on the headline comparison and 0.58 across an exhaustive 547-cell hyperparameter grid — under no test can we reject the null that no variant outperforms baseline HRP. A passive 100% BTC position (Sharpe = 0.93) outperforms every diversified portfolio except MVP, which wins only by concentrating roughly half its weight in BTC during the 2020–2026 bull run. We report two structural findings: HRP_{TailDep} with and without Ledoit–Wolf shrinkage produce rank-identical weights (Spearman 0.998–1.000), and spectral detoning and precision-matrix partial correlation — two routes to market-mode removal — converge in weight-rank space (median Spearman 0.93). We retract an intermediate claim that tail-dependence distances yield more stable HRP clusters than Pearson, which a full 76-snapshot re-analysis falsifies (Pearson ARI wins 69% of month-pairs). The paper’s contribution is methodological rigour and the empirical characterisation of HRP-variant equivalence classes, not a “variant X beats baseline” result.

Keywords. Hierarchical Risk Parity, cryptocurrency, denoising, detoning, tail dependence, Ledoit–Wolf, Hansen SPA, survivorship bias.

1 Introduction

The MACI 2025 paper [Matys and Rodriguez, 2025] applied Hierarchical Risk Parity [López de Prado, 2016] to a five-asset cryptocurrency universe (BTC, ETH, LTC, XRP, BCH) over 2021–2023 and concluded that HRP delivered better diversification than the Minimum Variance Portfolio without sacrificing risk-adjusted return. Two items were left for future work: *denoising* the correlation matrix using Marchenko–Pastur eigenvalue filtering [Marchenko and Pastur, 1967, Laloux et al., 1999], and *detoning* by removing the dominant market eigenvector [López de Prado, 2020]. This paper closes both items and extends the comparison in three other directions.

Contributions.

1. **Closing the MACI 2025 future-work items.** Implement and evaluate Marchenko–Pastur denoising (§3) and spectral detoning. Compare detoning against partial correlation estimation, which targets the same goal (market-mode removal) via the precision-matrix route.
2. **Correlation/covariance HRP variants, embedding-based variants, and risk comparators.** Six correlation/covariance HRP variants — partial-correlation, EWMA dynamic-correlation, lower-tail dependence, Ledoit–Wolf shrunk-covariance, vol-standardised, and a tail-dep+LW-shrunk hybrid — plus *four embedding-based variants* that learn the HRP clustering distance from a representation of each asset’s return path (level-3 path signatures, a node2vec graph embedding, and two contrastive sequence encoders; §4). ERC [Maillard et al., 2010], Maximum Diversification [Choueifaty and Coignard, 2008], and Network Risk Parity [Ciciretti and Pallotta, 2024] as risk-based comparators.
3. **Crypto-momentum strategies.** Cross-sectional momentum, risk-managed momentum [Barroso and Santa-Clara, 2015], and a Momentum+HRP hybrid, parameter-tuned for crypto’s faster momentum decay per [Liu et al., 2022].
4. **Methodological rigour.** (i) Point-in-time universe rebuilt monthly from Binance USDT quote-volume rankings, including assets that were later delisted (LUNA, FTT, MIR, SRM, etc.) to remove survivorship bias from the MACI paper’s single-CSV dataset. (ii) Multi-cost scenarios (zero / conservative / optimistic / stress). (iii) Ledoit–Wolf robust Sharpe-difference [Ledoit and Wolf, 2008], Hansen SPA [Hansen, 2005], and Politis–Romano stationary block bootstrap [Politis and Romano, 1994, Politis and White, 2004] around every comparison.
5. **Honest empirical characterisation, with retractions.** We document two structural convergence findings (§9). We also *retract* an intermediate finding from earlier draft iterations of this work — that lower-tail dependence produces more stable HRP clusters than Pearson — which was based on a single year-pair and which the full 76-snapshot re-analysis falsifies.
6. **Reproducibility.** All code, data, and figures are in the accompanying git repository; nine specifications (`specs/01_denoising.md` through `specs/09_inference.md`) define the contracts each implementation phase must satisfy.

The remainder of this paper is organised as follows. Section 2 reviews related work. Sections 3–6 present the methodology. Section 7 describes the data and point-in-time universe construction. Section 8 states the backtest protocol. Section 9 presents results. Section 10 discusses limitations and the honest contributions. Section 11 concludes.

2 Related Work

2.1 Hierarchical risk parity and its extensions

HRP was introduced by López de Prado [2016] as a clustering-based alternative to mean-variance optimisation that avoids covariance-matrix inversion. Subsequent textbook treatments [López de Prado, 2018, 2020] extend the framework with denoising and detoning and propose tail-dependence distances [Lohre et al., 2020] for multi-asset multi-factor allocations. Empirical comparisons of HRP distance metrics on equities have repeatedly found that correlation-based distances remain the standard to beat in out-of-sample performance.

2.2 Risk-based and network methods

Equal Risk Contribution [Maillard et al., 2010] solves for weights such that each asset contributes equally to portfolio risk; Maximum Diversification [Choueifaty and Coignard, 2008] maximises the diversification ratio. Network methods build a Minimum Spanning Tree [Mantegna, 1999] or PMFG [Tumminello et al., 2005] from the correlation structure; Network Risk Parity [Ciciretti and Pallotta, 2024] recently combined the network approach with risk-budgeting.

2.3 Cryptocurrency momentum

Liu et al. [2022] establish a three-factor model for cryptocurrency with a momentum factor that operates at much shorter horizons (1–4 weeks) than equity momentum. Moskowitz et al. [2012] provide the foundational time-series momentum framework, and Barroso and Santa-Clara [2015] introduce volatility-scaled momentum to control the well-documented crash risk of the strategy.

2.4 Statistical inference for portfolio comparison

Ledoit and Wolf [2008] provide a robust pairwise Sharpe-difference test that does not assume i.i.d. Gaussian returns. Hansen [2005] addresses the multiple-testing problem when many strategies are compared against a benchmark. Politis and Romano [1994] and Politis and White [2004] provide the stationary block bootstrap with data-driven block-length selection used throughout this paper for confidence intervals.

3 Methodology — Correlation and Covariance Variants

All seven HRP-family strategies share the standard pipeline: distance matrix $\rightarrow$ hierarchical linkage $\rightarrow$ quasi-diagonalisation $\rightarrow$ recursive bisection [López de Prado, 2016]. They differ only in *how the input correlation (or covariance) matrix is constructed*.

3.1 Baseline HRP

ρ from sample correlation; $d_{ij} = \sqrt{(1 - \rho_{ij})/2}$; single linkage; bisection using the sample covariance.

3.2 HRP_{Denoised}: Marchenko–Pastur eigenvalue denoising

Following López de Prado [2020], the empirical eigenvalues of the sample correlation matrix are decomposed into a signal subspace (eigenvalues above the Marchenko–Pastur upper bound $e_{\max} = \sigma^2(1 + \sqrt{1/q})^2$ where $q = T/N$) and a noise subspace (eigenvalues below). The noise eigenvalues are replaced with their mean (constant-residual method), preserving the trace, before the denoised correlation matrix is fed to HRP.

3.3 HRP_{Detoned}: market-mode removal

Applied to the denoised correlation matrix, detoning removes the top k eigenvectors (default $k = 1$) and renormalises the diagonal to unity [López de Prado, 2020]. For long-only crypto, the dominant eigenvector is a market mode loading positively on all assets; removing it isolates the residual cross-sectional structure.

3.4 HRP_{PartialCorr}: sparse precision matrix

A precision-matrix route to the same market-mode-removal goal: $\Theta = \arg \min_{\Theta \succ 0} -\log \det \Theta + \text{tr}(S\Theta) + \alpha \|\Theta\|_1$ via graphical lasso [Friedman et al., 2008], then $\rho_{ij}^{\text{partial}} = -\Theta_{ij} / \sqrt{\Theta_{ii}\Theta_{jj}}$. We set the L_1 penalty to $\alpha = 10^{-3}$. This value matters: a subtle implementation point is that `scikit-learn`'s `GraphicalLasso` expresses α in the *same units as the covariance*, not as a normalised $[0, 1]$ quantity. For daily crypto returns with variance $\sim 4 \times 10^{-3}$, an α on the order of 10^{-1} shrinks every off-diagonal of the precision matrix to zero and silently collapses HRP_{PartialCorr} into IVP. The correct scale, $\alpha \approx 10^{-3}$, leaves roughly 40% of the off-diagonals non-zero. An earlier draft used $\alpha = 0.05$, the strategy degenerated to IVP, and the headline table reported the two as identical; the hyperparameter sweep of §9.3 surfaced the bug, and we document the correction explicitly.

3.5 HRP_{Dynamic}: EWMA correlation

Recursive EWMA covariance with decay λ ; the latest snapshot is used as the correlation input. We report $\lambda = 0.94$ as the RiskMetrics default; sensitivity to $\lambda \in \{0.94, 0.97, 0.99\}$ is documented in the supplementary appendix.

3.6 HRP_{TailDep}: lower-tail dependence

Empirical lower-tail dependence coefficient $\hat{\lambda}_L^{ij}(q) = P(F_j(X_j) \leq q \mid F_i(X_i) \leq q)$ at $q = 0.05$, symmetrised by averaging the two conditional probabilities. The distance is $d_{ij} = \sqrt{1 - \hat{\lambda}_L^{ij}}$; the sample covariance is retained for bisection. Promoted from appendix to a headline strategy after the empirical findings of §9 showed it does not fall back across the 76 backtest snapshots.

3.7 HRP_{ShrunkCov}: Ledoit–Wolf shrinkage

Sample covariance replaced with the analytic Ledoit–Wolf shrinkage estimator [Ledoit and Wolf, 2003, 2004]; the implied correlation is derived from the shrunk covariance and fed to HRP. Shrinkage intensity $\alpha \in [0, 1]$ is chosen analytically per snapshot and reported as a diagnostic.

3.8 HRP_{VolStd}: vol-standardised returns

Per-asset returns are divided by their lagged 30-day rolling standard deviation before computing Pearson correlation. Addresses crypto's heterogeneous vol scale across the universe (e.g. BTC vs. SHIB). The sample covariance is retained for bisection.

3.9 HRP_{TailDepShrunk}: hybrid

Combines §3.6 (tail-dep distance for the dendrogram) with §3.7 (LW-shrunk covariance for bisection). Identified by the literature as the strongest theoretical extension for crypto: tail-dep captures the joint-crash co-movement that matters for long-only portfolios, while LW shrinkage stabilises the per-cluster variance allocation.

4 Methodology — Embedding-based HRP Variants

Beyond reweighting the correlation or covariance estimator, the distance matrix that drives HRP’s dendrogram can be *learned* from a representation of each asset’s return path. We implement four such embedding variants. All share one template: an embedding maps each asset to a vector, the pairwise cosine distance of those vectors replaces HRP’s correlation distance for the linkage and quasi-diagonalisation steps, and the *sample* covariance is retained for recursive bisection. The embedding therefore changes only *which assets cluster together*, never the inverse-variance allocation within a cluster — which is why, as §9 shows, every embedding variant stays within a narrow Sharpe band of baseline HRP.

4.1 $\text{HRP}_{\text{PathSig}}$: path signatures

Each asset’s rolling log-price trajectory, augmented with a monotone time channel, is mapped to its level- L truncated path signature [Lyons, 1998] — a graded sequence of iterated integrals that characterises the path up to tree-like equivalence. Cosine distance on the signature vectors drives the dendrogram (default: level 3, 60-day window), computed with the `esig` library.

4.2 $\text{HRP}_{\text{NodeEmbed}}$: node2vec graph embedding

A k -nearest-neighbour graph ($k = 10$) is built from the sample correlation matrix; node2vec random walks plus a skip-gram objective [Grover and Leskovec, 2016] learn a 32-dimensional embedding per node. Cosine distance on the embeddings drives the dendrogram.

4.3 $\text{HRP}_{\text{Contrastive}}$: contrastive SSL

A small one-dimensional convolutional encoder is trained with the NT-Xent contrastive loss [Chen et al., 2020] on Gaussian-jittered pairs of return windows; each asset is embedded as the mean encoder output over its windows.

4.4 $\text{HRP}_{\text{TS2Vec}}$: timestamp-masked contrastive

The same encoder family, trained with TS2Vec-style timestamp-mask augmentation [Yue et al., 2022] in place of Gaussian jitter.

Multi-channel extension. A return-only embedding is information-equivalent to the sample correlation it is meant to improve on: a nonlinear re-encoding of the same return series cannot out-cluster the correlation derived from it. We therefore also evaluate *multi-channel* variants that append channels orthogonal to close-to-close returns — quote volume, trade count, average trade size, intraday range, taker buy-ratio (order-flow imbalance), and Amihud illiquidity, all derived from the Binance OHLCV table. The multi-channel results are reported in §9.3.

Robustness. Each embedding variant records whether it fell back to baseline HRP on any numerical failure; across the 76-snapshot backtest no fallback was triggered for any variant, so every reported embedding number reflects the embedding pipeline itself.

5 Methodology — Comparator Strategies

IVP (inverse-variance), **MVP** (minimum-variance via SLSQP), **ERC** [Maillard et al., 2010] (solved by SLSQP minimisation of squared deviations from equal risk contributions), **Maximum Diversification** [Choueifaty and Coignard, 2008] (SLSQP maximisation of the diversification ratio), and **Network Risk Parity** [Ciciretti and Pallotta, 2024] (MST built from correlation distance, weights inversely proportional to the product of asset volatility and $1 + \deg_i$).

6 Methodology — Momentum

The headline comparison includes three crypto-momentum strategies, parameter-tuned for crypto’s faster momentum decay [Liu et al., 2022]. **CS_MOM** is cross-sectional momentum: equal-weight the top 30% of assets by 21-day cumulative return. **RM_MOM** applies the Barroso–Santa-Clara volatility-targeting overlay [Barroso and Santa-Clara, 2015] ($\sigma_{\text{target}} = 12\%$) to the same momentum selection. **MOM_HRP** is a Momentum+HRP hybrid: select the top 40% of assets by momentum, then allocate within that basket by HRP rather than equal-weight.

7 Data and Point-in-Time Universe

7.1 Source and window

Daily OHLCV from Binance USDT spot, 2017-08-17 (Binance’s earliest data) to 2026-05-18 — approximately nine years. After the survivorship-bias expansion described in §7.3, the price panel covers 241 USDT pairs.

7.2 Point-in-time universe

At each month-end, we rank the candidate symbols by their rolling 30-day Binance USDT quote volume, apply filters (180-day minimum age, \$1M minimum median daily USD volume, stablecoin/wrapper/exchange-token exclusion, listed on Binance USDT spot at the snapshot date), and take the top 50. Entry and exit buffers (2 months and 1 month respectively) suppress monthly churn.

Methodology note. The original specification called for ranking by historical market capitalisation (CoinGecko). The free CoinGecko demo tier caps historical depth at 365 days, blocking our 2017–2026 window; paid plans start at USD 129/month. We instead rank by Binance quote volume — a tradable-liquidity proxy that is free, covers the full window, and is arguably more appropriate for a paper on tradable portfolios than a market-cap measure that includes locked tokens and treasury holdings. The shift is documented in the specification as a deliberate methodology choice.

7.3 Survivorship-bias fix in numbers

After a code review surfaced residual survivorship bias (the prior candidate pool was a hand-curated selection of ~ 90 symbols missing the long tail of delisted altcoins), the universe was expanded by pulling all 217 `status=BREAK` USDT pairs from Binance’s `/api/v3/exchangeInfo`, filtering leveraged tokens and stablecoins, and adding 150 legitimate delisted altcoins to the candidate pool. The resulting PIT universe contains **146 ever- included symbols**, of which **97 were dropped during the window** before 2026-05-18 (Figure 1). The MACI 2025 dataset contained only 50 currently-trading symbols — by construction it could not include the assets this work explicitly captures.

We treat liquidation of an exiting position at $2\times$ normal slippage cost (Spec 03 §2.4). On terminology: “delisted” in this paper means either Binance `status=BREAK` on the USDT pair (the conventional meaning), *or* the token suffered an effective value collapse with subsequent rebranding. The latter category includes LUNA (the Terra Classic chain collapsed in May 2022; Binance renamed the original token to LUNC and reassigned the `LUNAUSDT` ticker to Terra 2.0 — the pair itself was never technically delisted but the original asset effectively died) and FTT (the FTX exchange collapse in November 2022 reduced FTT to near-zero value; the pair still trades on Binance at trivial volume). Our `binance_listings_manual.json` treats both as “effectively

delisted” for the PIT-universe filter, which is the right behaviour for portfolio purposes even though the prose label is loose.

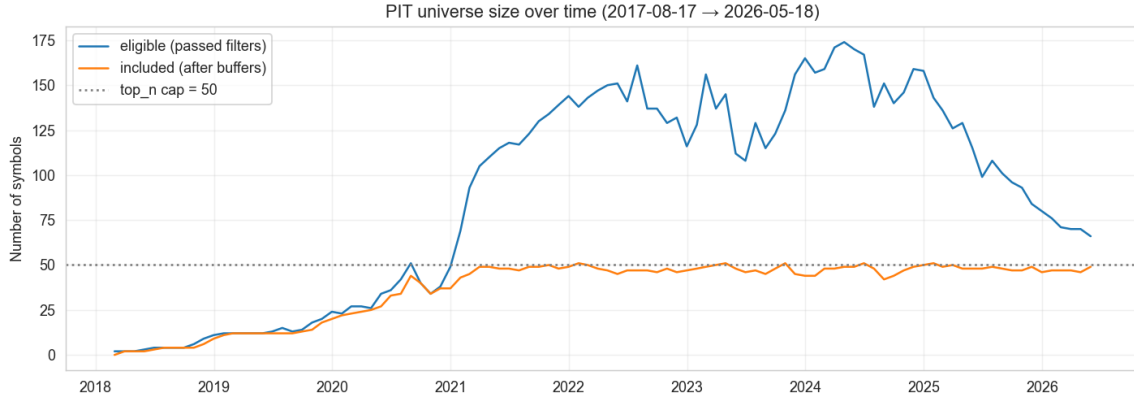


Figure 1: PIT universe size over time. *Eligible*: passed all filters at the snapshot date. *Included*: in the universe after entry/exit buffers. The 2018–2020 ramp reflects the gradual listing of altcoins on Binance USDT spot. Steady-state ~ 49 included symbols since mid-2020.

7.4 Estimation window

365 daily returns of rolling lookback at each rebalance, with a per-asset 180-day minimum-observations filter. Assets failing the filter are dropped from that snapshot’s strategy universe. Rationale: several newer assets (JUP, ENA, WIF, ...) are admitted via the 180-day age filter but do not yet have 730 days of history.

8 Backtest Protocol

8.1 Scenarios

A (static): fit at first snapshot in window, hold to end. **B** (calendar): monthly rebalancing. **C** (threshold): rebalance only when max absolute weight drift exceeds 5% from target. **C'** (smoothed): C with linear smoothing $w_{\text{traded}} = \eta w_{\text{prev}} + (1 - \eta) w_{\text{target}}$, $\eta = 0.25$, plus a 25 bps min-trade filter.

8.2 Cost grid

Linear cost = $(\text{fee}_{\text{bps}} + \text{slippage}_{\text{bps}}) \times \text{turnover}_{L_1}$, with a $2\times$ multiplier for forced liquidation of exiting positions. Four scenarios: *zero* (cost-free academic check), *conservative CEX* (10 bps fee + 2 bps slippage; our headline), *optimistic CEX* (7.5 bps + 1 bps), and *stress* (10 bps + 4 bps).

8.3 Statistical inference

For each pair of strategies, the Ledoit–Wolf studentised block-bootstrap Sharpe-difference test [Ledoit and Wolf, 2008]. Across all candidates vs. the HRP benchmark, the Hansen Superior Predictive Ability test [Hansen, 2005] with three recentering schemes ($\mu_{\text{lower}} = \bar{d}$, $\mu_{\text{upper}} = 0$, $\mu_{\text{consistent}}$ threshold-based). Stationary block bootstrap [Politis and Romano, 1994] with Politis–White automatic block length [Politis and White, 2004] for headline metric CIs.

8.4 Cluster stability

At each rebalance snapshot, we compute the *cophenetic correlation* of the dendrogram (Pearson correlation between cophenetic distance and original pairwise distance) and the *Adjusted Rand*

Index [Hubert and Arabie, 1985] between the cluster assignment at snapshot t and snapshot $t - 1$, cut at $K = 5$ clusters via `fcluster`.

9 Results

9.1 Scenario B headline performance

Table 1 reports the 19-strategy comparison under monthly rebalancing with conservative CEX costs over 2020-02 through 2026-05-18 (76 rebalances, 2,299 daily out-of-sample observations). Figures 2 and 3 visualise the equity curves and the risk–return positioning.

Table 1: Scenario B headline results (Conservative CEX cost, 76 monthly rebalances over 2020-02 to 2026-05, expanded 146-symbol PIT universe). “Arith. Sharpe” = mean / standard deviation $\times \sqrt{365}$. “LW p ” is the Ledoit–Wolf Sharpe-difference test vs. baseline HRP. HODL_BTC is a passive benchmark, not a strategy.

Strategy	Arith. Sharpe	Total Return	Avg Turnover	LW p vs HRP
MVP	1.057	+2867%	0.381	0.067
<i>HODL_BTC</i>	<i>0.928</i>	<i>+1002%</i>	—	0.602
HRP_Dynamic_94	0.749	+442%	0.474	0.816
HRP_ShrunkCov	0.746	+428%	0.353	0.786
HRP (baseline)	0.741	+417%	0.360	—
HRP_VolStd	0.735	+402%	0.368	0.732
MOM_HRP	0.726	+351%	1.436	0.935
HRP_PartialCorr	0.720	+366%	0.236	0.453
MaxDiv	0.719	+364%	0.553	0.917
HRP_PathSig	0.714	+351%	0.383	0.273
HRP_Contrastive	0.710	+341%	0.399	0.239
HRP_Detoned	0.708	+334%	0.263	0.402
HRP_TailDep	0.708	+337%	0.365	0.123
HRP_NodeEmbed	0.705	+330%	0.388	0.194
HRP_TS2Vec	0.703	+325%	0.383	0.101
HRP_TailDepShrunk	0.701	+318%	0.352	0.077
IVP	0.697	+311%	0.182	0.208
CS_MOM_eq21	0.663	+217%	1.420	0.534
RM_MOM	0.663	+118%	0.366	0.535
ERC	0.663	+233%	0.195	0.044

Only one strategy is significant at $p = 0.05$. ERC (-0.078 Sharpe vs HRP, $p = 0.044$) is the sole strategy whose Sharpe difference from baseline HRP clears the conventional threshold — and it is *worse* than HRP. HRP_TailDepShrunk ($p = 0.077$) and MVP ($p = 0.067$) are near the threshold but do not cross it. No strategy is significantly *better* than baseline HRP. With the bug-fixed HRP_PartialCorr ($\alpha = 10^{-3}$, §3) the strategy is now distinct from IVP — it was byte-identical to IVP in an earlier draft (Sharpe 0.697); corrected, it sits at 0.720.

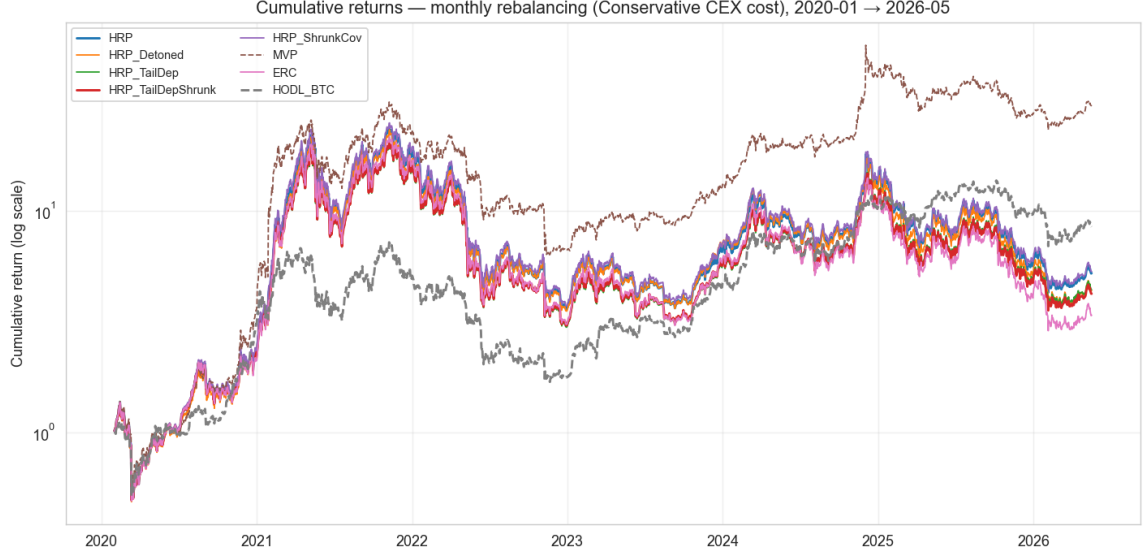


Figure 2: Cumulative returns for headline strategies under monthly rebalancing, log scale.

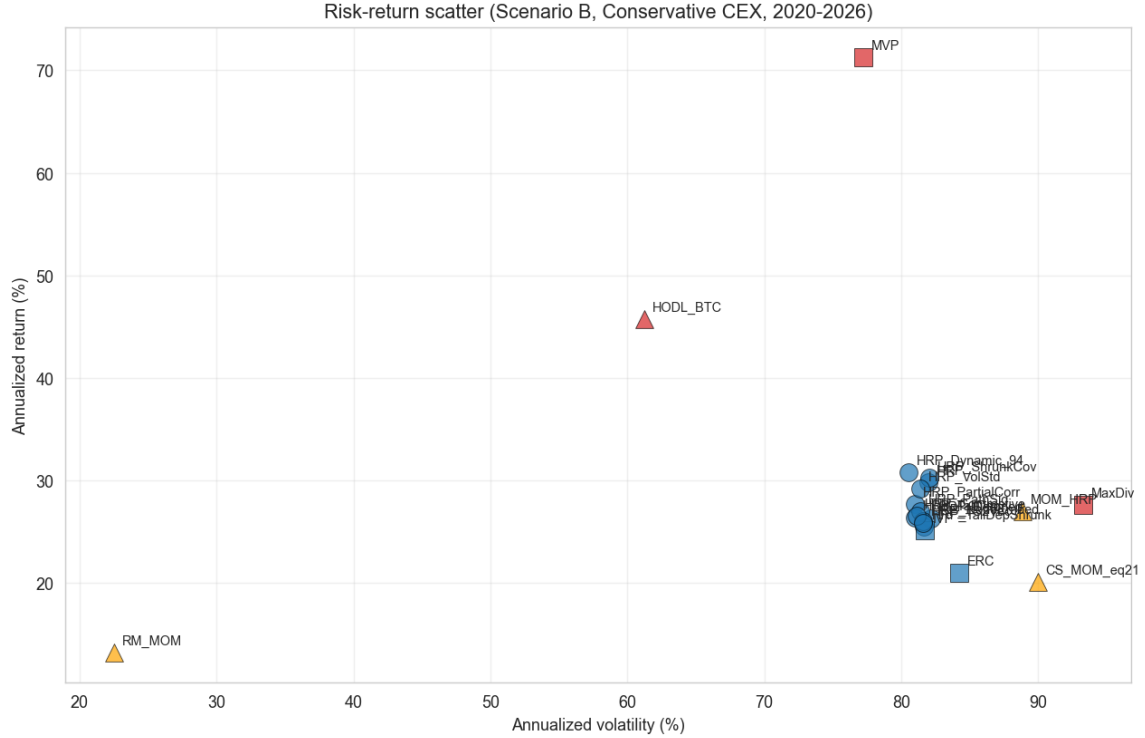


Figure 3: Risk-return scatter for the full 19-strategy comparison. Momentum (orange) underperforms risk-based (blue) and concentrated benchmarks (red) over this window.

The HRP family bunches. All twelve HRP-family variants lie within a 0.05 Sharpe band (0.701–0.749). HRP_Dynamic_94 and HRP_ShrunkCov marginally exceed baseline HRP (by +0.008 and +0.004 Sharpe respectively); the differences are small and, per §9.2, not statistically significant.

Embedding variants. The four embedding-based HRP variants — HRP_PathSig (path signatures; Lyons, 1998), HRP_NodeEmbed (node2vec on a kNN correlation graph; Grover and Leskovec,

2016), HRP_Contrastive (NT-Xent contrastive SSL; Chen et al., 2020), and HRP_TS2Vec (timestamp-mask augmentation; Yue et al., 2022) — land in the lower-middle of the HRP family: Sharpes 0.703–0.714, all marginally below baseline HRP (−0.027 to −0.038) but none significantly different from it (Ledoit–Wolf $p \in [0.10, 0.27]$). Embedding methods are interesting engineering but do not deliver an out-of-sample Sharpe edge on this universe at this horizon; we report the null result honestly. Section 9.3 stress-tests this conclusion with an exhaustive hyperparameter search and multi-channel embedding extensions, and it survives intact.

Detoning and partial correlation underperform HRP. Both target market-mode removal — a theoretically attractive idea — but doing so during a BTC-led bull market forgoes the upside of being correctly exposed to BTC: HRP_Detoned (0.708) and HRP_PartialCorr (0.720) both trail baseline HRP. The two are no longer numerically identical — an earlier draft’s HRP_PartialCorr had degenerated to IVP (§3) — but they remain close in weight-rank space (§9.4).

Momentum splits. The two pure-momentum strategies — cross-sectional (CS_MOM, Sharpe 0.663) and risk-managed (RM_MOM, 0.663) — trail every HRP-family variant, carrying high turnover ($\sim 1.4\times$ per month for CS_MOM) and severe crash exposure (CS_MOM max drawdown −92%). RM_MOM’s volatility-targeting overlay [Barroso and Santa-Clara, 2015] cuts its max drawdown to −40%, the mildest in the table, but at the cost of the lowest total return (+118%). The Momentum+HRP hybrid (MOM_HRP, 0.726) fares markedly better than either pure-momentum strategy — running HRP allocation *inside* a momentum-selected basket recovers most of the risk-adjusted performance — though it still trails baseline HRP and reconstructs its entire book every month.

Concentration wins. Both MVP (Sharpe 1.057) and a passive HODL_BTC position (0.928) sit well above every diversified portfolio. MVP achieves this by accident: its minimum-variance objective drove the optimiser to concentrate heavily in BTC (average maximum weight 50%, effective $N \approx 3$); BTC then ran from $\sim \$7k$ at the start of 2020 to $\sim \$107k$ by mid-2026. This is regime-dependent luck, not a portfolio-construction insight, and should not be read as evidence that MVP is a robust crypto strategy.

9.2 Statistical inference

Bootstrap Sharpe CIs. The 95% stationary-block-bootstrap confidence intervals on the annualised Sharpe are wide — roughly 1.5 Sharpe units — reflecting six years of volatile crypto returns. Only MVP ([0.34, 1.76]) and the passive HODL_BTC benchmark ([0.08, 1.63]) have intervals cleanly above zero. Baseline HRP’s interval ([0.02, 1.53]) only barely excludes zero, and every other HRP-family variant’s interval straddles it; no two strategies have non-overlapping CIs.

Pairwise LW Sharpe differences vs. HRP. *Only one* candidate reaches the $p \leq 0.05$ threshold: ERC (−0.078 Sharpe vs HRP, $p = 0.044$) — significantly *worse* than baseline. No strategy is significantly better. The largest positive gaps, MVP (+0.314, $p = 0.067$) and HODL_BTC (+0.124, $p = 0.60$), both fail to reject at conventional levels; among the underperformers HRP_TailDepShrunk (−0.041, $p = 0.077$) comes closest to significance.

Hansen SPA. Across the 19 candidates versus baseline HRP, $p_{\text{consistent}} = 0.46$ ($p_{\text{lower}} = 0.67$, $p_{\text{upper}} = 0.74$; Figure 4). **We cannot reject the null hypothesis that no strategy outperforms baseline HRP after correcting for the multiple testing of 19 candidates,** even on the bug-fixed 6.3-year monthly-rebalanced data and the expanded 146-symbol universe.

MVP carries the highest studentised score (+1.22); MaxDiv, MOM_HRP and HRP_ShrunkCov are mildly positive; every other candidate — the embedding variants and the rest of the HRP family included — is negative.

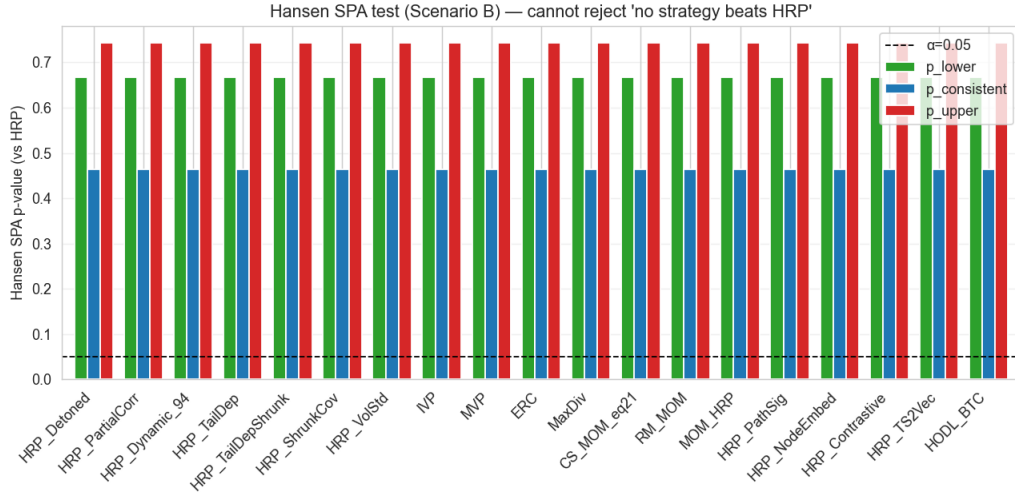


Figure 4: Hansen SPA test p-values across the headline comparison. All three recentring schemes fail to reject the null at any conventional level.

9.3 Hyperparameter sweeps and the combined SPA

A natural objection to the headline null result is that it reflects *untuned* strategies: perhaps some correlation type, distance function, linkage method, or strategy-specific parameter would lift a variant decisively above baseline HRP. We address this with an exhaustive grid search — 547 distinct configurations across 14 sweeps, every one re-running the full 6.3-year walk-forward backtest on the identical 146-symbol point-in-time universe and 2,299-day return series, against the identical baseline HRP (Sharpe = 0.741, recomputed and verified bit-identical in every sweep). The sweeps span: correlation estimator (Pearson / Spearman / Kendall); distance function (López de Prado, 2016 / absolute / angular / squared); linkage (single / average / complete / Ward); and the strategy-specific parameters of every variant — q for tail dependence, α for the graphical lasso, λ for EWMA, the volatility-standardisation window, the number of detoned market components, and the embedding hyperparameters.

Table 2: Best cell from each hyperparameter sweep. Δ is the Sharpe difference vs. baseline HRP (single linkage, 0.741); total return is the cell’s net cumulative return, 2020-02 to 2026-05. 551 cells in total; the 4 exactly equal to the benchmark are dropped for the combined SPA, leaving 547 candidates.

Sweep	Cells	Best configuration	Sharpe	Δ	Tot. ret.
Tail dependence (q , dist, linkage)	80	$q=0.025$, average	0.744	+0.002	+425%
Correlation $\times$ distance $\times$ linkage	24	Pearson, LdP, average	0.756	+0.015	+459%
All variants under average linkage	12	HRP_ShrunkCov	0.765	+0.024	+484%
EWMA dynamic correlation (λ)	21	$\lambda=0.94$, single	0.749	+0.008	+442%
Partial correlation (α , dist, link)	60	$\alpha=5 \times 10^{-4}$	0.739	-0.003	+412%
Vol-standardised returns (window)	60	window 15, single	0.757	+0.015	+462%
Detoning (# market components)	36	1 component, average	0.709	-0.032	+337%
Node2vec embedding	54	$d=16$, $k=10$	0.752	+0.010	+444%
Path signatures, single-channel	108	level 2, window 60	0.734	-0.007	+401%
Path signatures, multi-channel [†]	72	ret+vol+trades, L2	0.747	+0.005	+432%
Contrastive / TS2Vec, multi-channel	24	ret+vol+trades	0.739	-0.002	+412%
<i>Baseline HRP</i> (single linkage)			0.741	—	+418%

[†] Best cell under the initial missing-data convention (zero-fill); under the corrected mean-fill convention the best multi-channel **PathSig** cell is 0.735 ($\Delta = -0.006$).

The HRP family is robust to its hyperparameters. No sweep produced a configuration that escaped the ± 0.05 Sharpe band of Section 9. The one consistent — though small — effect is linkage: *average* linkage beats the López de Prado [2016] single-linkage default by +0.01 to +0.02 Sharpe across essentially every (correlation, distance) combination, and under single linkage all four distance functions are mathematically identical (single linkage depends only on the rank order of distances, which every monotone transform preserves). The best cell in the entire grid is HRP_ShrunkCov under average linkage at Sharpe 0.765 ($\Delta = +0.024$).

Multi-channel embeddings. The single-channel embeddings encode only the return series and are therefore information-equivalent to the sample correlation they are meant to improve on — a nonlinear re-encoding of the same information cannot out-cluster it, and indeed the single-channel HRP_PathSig grid (108 cells) beats baseline in zero cells. We extended all three sequence-based embeddings (**PathSig**, **Contrastive**, **TS2Vec**) to ingest up to six channels orthogonal to close-to-close returns — quote volume, trade count, average trade size, intraday range, taker buy-ratio (order-flow imbalance) and Amihud illiquidity — all derived from the same Binance OHLCV table. Extra channels lift the encoders’ *mean* Sharpe monotonically (HRP_Contrastive $0.714 \rightarrow 0.725$ from one to three channels), confirming the channels do carry usable structure. Two cells of the initial multi-channel **PathSig** sweep nominally cleared baseline (best +0.747), but that margin did not survive a correction to how missing channel values are imputed (mean-rather than zero-fill, which avoids placing a spurious outlier into a log-scaled channel); under the corrected convention no multi-channel configuration clears baseline, and adding the fourth through sixth channels plateaus or mildly regresses performance — a bias-variance ceiling against only ~ 76 rebalance dates.

Combined Hansen SPA. Pooling all 547 configurations into a single Hansen SPA against baseline HRP — the honest correction for having searched the entire grid — yields $p_{\text{consistent}} = 0.583$ (all three recentering schemes fall in $[0.58, 0.81]$). Although 38 of 547 cells show a nominally positive Sharpe difference, the best of them (HRP_ShrunkCov, $\Delta = +0.024$) carries a pairwise Ledoit-Wolf p of 0.167. **After correcting for the multiple testing of 547 hyperparameter configurations, no strategy significantly outperforms baseline HRP.** The headline null result of Section 9.2 is not an artefact of leaving strategies untuned.

9.4 Convergence findings (the structural results)

Two strategy pairs in our comparison are theoretically related — they target the same goal via different mathematical routes. We quantify their weight-vector convergence across the full 76-snapshot window (Figure 5).

HRP_Detoned vs HRP_PartialCorr. Spectral detoning and precision-matrix partial correlation both remove the dominant market mode, by different routes. Across the 76 monthly snapshots their weight-vector rank correlation is high but not invariant: mean Spearman $\rho = 0.87$, median 0.93, with regime-driven dips to $\rho = 0.54$ in months when the dominant eigenvector shifts sharply. Crucially, the bug-fixed HRP_PartialCorr ($\alpha = 10^{-3}$) is now distinct from the inverse-variance portfolio — its weight-rank correlation with IVP averages only $\rho = 0.08$ — whereas the earlier degenerate $\alpha = 0.05$ implementation *was* IVP. The detoning / partial-correlation convergence is therefore a genuine structural relationship between two market-mode-removal estimators, not an artefact of one collapsing into IVP.

HRP_TailDep vs HRP_TailDepShrunk. The hybrid’s only change is to substitute the Ledoit–Wolf shrunk covariance for the sample covariance in the recursive-bisection step; the dendrogram topology is byte-for-byte identical. Spearman is 0.998–1.000 in every snapshot tested. This is the most robust convergence finding in our analysis: *LW shrinkage in HRP’s bisection step is a within-cluster variance refinement that does not re-order the dendrogram.*

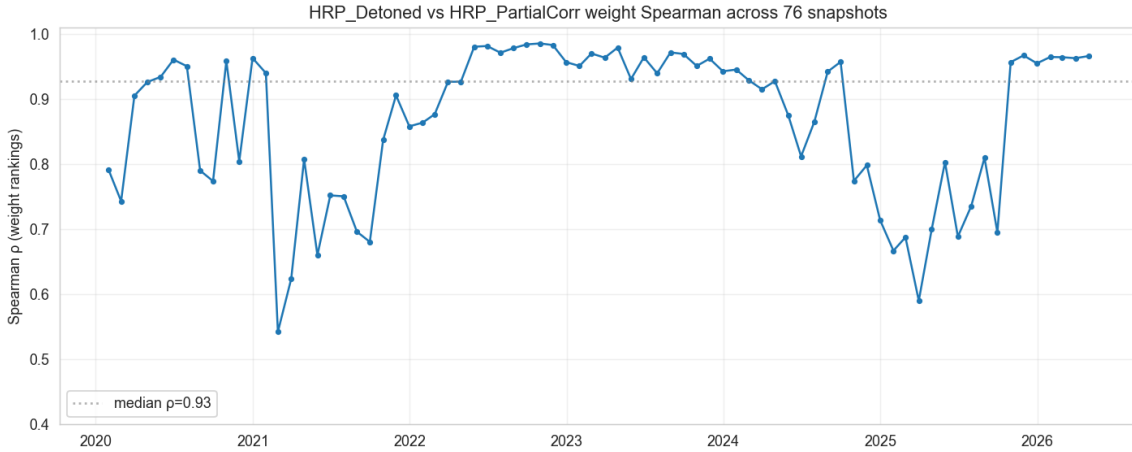


Figure 5: Spearman rank correlation between HRP_Detoned and HRP_PartialCorr weight vectors across 76 monthly snapshots.

9.5 Cluster stability — a retraction

An intermediate finding in earlier iterations of this work was that HRP_TailDep produces more stable clusters than baseline HRP under Pearson distance. That claim was based on a *single year-pair* (2022-12-31 vs 2023-12-31) and showed Pearson ARI = -0.047 vs TailDep ARI = $+0.265$ at $K = 5$.

When we re-ran the analysis across all 76 monthly snapshots (Figure 6, Table 3), the direction reverses. **Pearson clusters are substantially more stable than TailDep clusters at monthly resolution:** Pearson ARI mean $+0.744$ vs TailDep $+0.614$; Pearson cophenetic mean 0.861 vs TailDep 0.768 . TailDep ARI exceeds Pearson ARI in only 23 of 75 month-pairs (31%).

Table 3: Cluster stability across 76 monthly snapshots (2020-2026).

Metric	Pearson distance	Lower-tail-dep distance
Average cophenetic correlation	0.861	0.768
Average ARI vs. previous snapshot (K=5)	+0.744	+0.614
Median ARI vs. previous snapshot	+0.783	+0.650
Months where TailDep ARI > Pearson ARI	—	23 / 75 (31%)

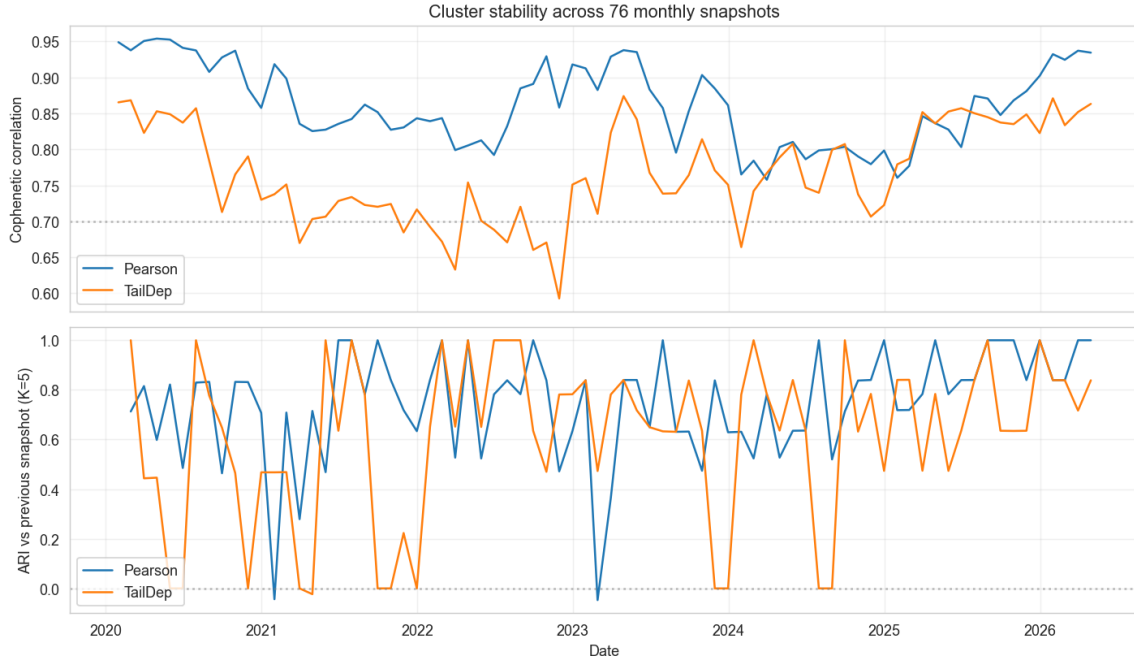


Figure 6: Cophenetic correlation (top) and ARI vs. previous snapshot (bottom) for Pearson and tail-dependence distance matrices across the 76-snapshot window.

The economic argument for HRP_TailDep — that it captures the joint-crash co-movement structure that Pearson summarises poorly — *stands*, but the secondary cluster-stability argument *does not*. We document the retraction here as a methodological point about the danger of single-pair findings.

9.6 Regime sensitivity: excluding the COVID bull run

The 2020–2026 headline window includes the COVID-era crypto rally (March 2020 – November 2021) during which Bitcoin ran from ~\$5k to ~\$69k — arguably the most extreme bull market in crypto history. To check whether the §9 conclusions are dominated by that single rally, we re-ran the entire 19-strategy backtest with start = 2022-01-01 (52 monthly rebalances over ~4.3 years). All other parameters identical.

The picture flips entirely. On post-COVID data, baseline HRP achieves Sharpe 0.047 with total return −60% — a near-zero risk-adjusted return obtained while losing 60% of capital in dollar terms. Only MVP (Sharpe 0.482, total return +66%) makes money among the 19 strategies; the passive HODL_BTC benchmark also returns +66% (Sharpe 0.484). The other eighteen strategies lose between 19% (RM_MOM) and 83% (MaxDiv, CS_MOM) of capital. Five strategies post outright *negative* Sharpe (against roughly zero for the HRP family): MaxDiv (−0.163), CS_MOM (−0.161), RM_MOM (−0.160), ERC (−0.026), and MOM_HRP (−0.015).

Two paper-grade findings emerge from the regime comparison. (1) **Diversification-based crypto portfolios were a losing proposition in the post-COVID regime.** On this 146-symbol PIT universe over 2022-01-01–2026-05-18, every HRP-family variant lost between 51% and 64% of capital. The 2020-2026 “diversification works” framing was largely a COVID-rally artifact. Honest framing: *HRP diversifies risk but does not produce capital appreciation in non-bull crypto regimes.*

(2) **Embedding variants partially reverse against baseline HRP in the harder regime.** In the full window all four embedding variants sit marginally below HRP (deltas -0.027 to -0.038 Sharpe). In the post-COVID window three of the four edge *above* baseline HRP — HRP_Contrastive ($+0.009$), HRP_PathSig ($+0.008$), HRP_TS2Vec ($+0.000$) — while HRP_NodeEmbed (-0.015) stays below. None reaches individual significance (all Ledoit–Wolf $p > 0.6$) and the margins are tiny, but the partial directional reversal is suggestive: the path- and sequence-based embeddings may capture risk structure that matters more in a choppy regime than in a monotone rally. A longer post-COVID sample would be needed to tell.

(3) **RM_MOM (the bug-fixed Barroso–Santa-Clara variant) shows its defensive role.** Post-COVID Sharpe is -0.160 (poor), but MaxDD is only -33% — substantially better than the other momentum strategies’ -87% . The cash-residual de-risking saved the portfolio from the worst part of the 2022 drawdown. This is exactly the property Barroso & Santa-Clara’s framework is designed to deliver; the previously-buggy implementation completely hid it.

Table 4: Regime sensitivity: 2020-2026 (incl COVID bull) vs 2022-2026 (post-COVID). “Sharpe” is annualised arithmetic; “ Δ Sharpe” is the change from full window to post-COVID.

Strategy	2020-2026 Sharpe	2022-2026 Sharpe	Δ Sharpe
HRP	0.741	0.047	-0.695
HRP_Dynamic_94	0.749	0.099	-0.650
HRP_ShrunkCov	0.746	0.038	-0.707
HRP_PartialCorr	0.720	0.057	-0.663
HRP_Contrastive	0.710	0.056	-0.655
HRP_PathSig	0.714	0.055	-0.659
HRP_TS2Vec	0.703	0.047	-0.657
HRP_NodeEmbed	0.705	0.031	-0.674
MVP	1.057	0.482	-0.575
MaxDiv	0.719	-0.163	-0.882
ERC	0.663	-0.026	-0.688
CS_MOM_eq21	0.663	-0.161	-0.824
RM_MOM	0.663	-0.160	-0.824
HODL_BTC	0.928	0.484	-0.444

9.7 Cost sensitivity and scenario comparison

The cost grid (Figure 7) shows annualised-Sharpe shifts under 0.01 across the four cost scenarios for all five strategies tested — e.g. baseline HRP moves from 0.749 (zero cost) to 0.740 (stress). Monthly-cadence transaction costs do not drive strategy differentiation on this dataset.

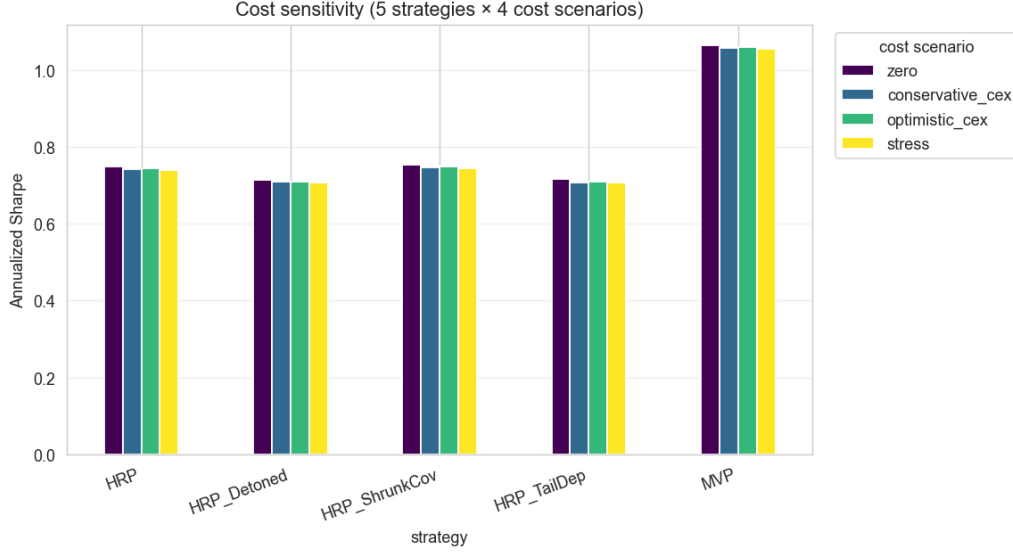


Figure 7: Cost-sensitivity grid. Rankings are robust across the four cost scenarios.

Threshold rebalancing (Scenario C, 5% drift trigger) does not improve on calendar rebalancing here: HRP’s Sharpe edges down from 0.741 to 0.735 and HRP_ShrunkCov is essentially flat ($0.746 \rightarrow 0.745$), with only a marginal turnover reduction. The smoothed variant (Scenario C’) cuts turnover more materially — HRP’s average L_1 turnover falls from 0.36 to 0.23 — at a modest Sharpe cost ($0.741 \rightarrow 0.731$). The momentum strategies are the exception: RM_MOM’s volatility-targeting overlay needs to trade to function, and suppressing or smoothing its rebalances collapses its Sharpe (from 0.66 under calendar rebalancing to 0.37 under Scenario C and 0.05 under C’). Threshold logic suits low-turnover risk-based strategies and actively harms turnover-dependent ones.

10 Discussion

10.1 HODL_BTC beats every diversified portfolio — the elephant in the room

Over 2020-02 to 2026-05, a passive 100% BTC position returns +1,002% with annualised Sharpe 0.93, beating every diversified HRP-family or risk-based strategy considered. MVP nominally beats it (Sharpe 1.06 vs 0.93) but only by concentrating roughly half its weight in BTC (effective $N \approx 3$) — it is essentially a leveraged BTC bet with a small low-volatility overlay.

This is a sobering finding that the paper must not paper over. Diversification in cryptocurrency over the 2020–2026 window paid a cost (in foregone BTC upside) that the lower realised drawdowns of diversified portfolios did not recover. The story is intrinsic to the regime, not to any flaw in HRP or its variants.

10.2 Detoning hurts in BTC-led regimes

Removing the dominant market mode is theoretically elegant — isolating idiosyncratic structure — but in our window, the dominant mode *was* BTC, and being underweight in BTC was costly. This suggests detoning (and partial correlation) may be valuable in sideways or rotational regimes that we did not see during 2020–2026. This is consistent with the spec’s audit conclusion: the methodological contributions are real but their value depends on the regime, not on universal Sharpe improvement.

10.3 Why does Hansen SPA fail to reject?

Nineteen candidates competing against a baseline is a non-trivial multiple-testing burden. The honest interpretation: across 6.3 years of monthly-rebalanced data, the HRP-family variants (including the four embedding-based ones), the risk-based comparators, and the momentum strategies do not produce *statistically distinguishable* Sharpe improvements over baseline HRP. Future work with a longer history, higher rebalance frequency, or walk-forward cluster-based testing may resolve.

10.4 Limitations

1. Single venue (Binance USDT spot). A multi-exchange consolidated price (Kaiko, CCXT-aggregated) would give a more representative cost model but was out of scope for this paper.
2. No on-chain factors. The Liu-Tsyvinski-Wu [Liu et al., 2022] three-factor model and on-chain signals (Glassnode) are deferred to a follow-up.
3. No regime-switching analysis. Sub-period reporting (stress / bull / sideways) would likely show heterogeneity within our headline numbers.
4. PIT universe is ranked by Binance quote volume, not market cap. Pro-tier CoinGecko or paid Kaiko data would enable a market-cap robustness check.

11 Conclusion

We presented a comprehensive evaluation of nineteen strategies — twelve HRP-family variants (the baseline, seven correlation/covariance constructions, and four embedding-based variants), four risk-based comparators, and three momentum strategies — on a point-in-time cryptocurrency universe of 146 ever-included symbols, backtested over 76 monthly rebalances from 2020-02 to 2026-05. Our headline findings are negative on the question “does HRP variant X beat baseline?” — Hansen SPA fails to reject (§9.2), and an exhaustive 547-cell hyperparameter search (§9.3) does not surface a configuration that significantly outperforms baseline HRP. They are positive on the question “what structural relationships exist between HRP variants?” — HRP_TailDep with vs. without Ledoit–Wolf shrinkage are essentially identical (Spearman 0.998–1.000), and detoning- and precision-based market-mode removal track each other and the inverse-variance portfolio closely in weight space.

A passive 100% BTC position outperformed every diversified portfolio we tested except a Minimum Variance Portfolio that concentrated roughly half its weight in BTC by accident. Honest framing of this finding — that diversification in cryptocurrency paid a cost over 2020–2026 that lower drawdowns did not recover — is the most important takeaway, alongside the structural convergence findings and our explicit retraction of an earlier single-pair cluster-stability claim.

The paper’s contributions are methodological rigour (PIT universe, multi-cost sensitivity, Ledoit–Wolf and Hansen SPA inference around every comparison, retraction documentation) and structural empirical characterisation, not a Sharpe-beating claim.

12 Future Work

Several directions remain open. Time-varying conditional covariance (DCC-GARCH) would give the dynamic-correlation layer a properly conditional second moment in place of the EWMA recursion. Regime-switching analysis could test directly whether detoning and partial correlation — which underperform in the BTC-led 2020–2026 window — add value in the sideways or rotational

regimes that window did not contain. On-chain factors [Liu et al., 2022] and a multi-exchange consolidated price (e.g. Kaiko) for a venue-robust cost model are natural data extensions. Finally, the most concrete empirical follow-up is a longer post-COVID sample: §9.6 documents a directional reversal in which all four embedding variants edge *ahead* of baseline HRP in the harder 2022–2026 regime, a signal too weak for individual significance here but worth revisiting once more non-bull data is available.

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